Labo 3 - ARN

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20.03.2026

First experience

Pretreatment

We chose the 25 lowest frequencies going from 1Hz to 25Hz, they cover all EEG bands that are relevent for sleep calssification. Frequencies that are higher carry only little discriminative infos for this task.

Using the `StandardScaler` centers each of the 25 features to mean 0 and std 1, thus stabilizing and speeding up training.

Architecture

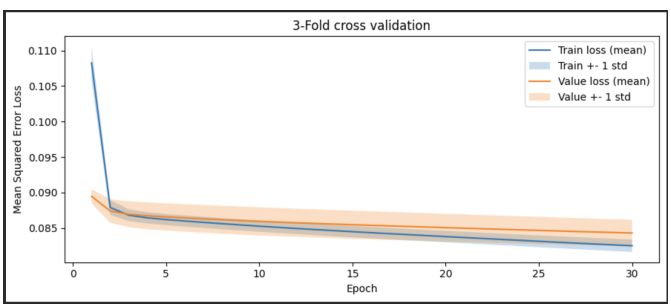
Model: "sequential_15"

Layer (type)	Output Shape	Param #
dense_38 (Dense)	(None, 16)	416
dense_39 (Dense)	(None, 1)	17

Total params: 433 (1.69 KB)
Trainable params: 433 (1.69 KB)
Non-trainable params: 0 (0.00 B)

We only use one hidden layer with 16 neurones as we only have 25 inputs to treat. Adding hidden layers and neurones would risk overfitting without having any benefits. The sigmoid output produces a probability between 0 and 1, and the threshold of 0.5 determines the predicted class. We also added a momentum of 0.9 so we avoid getting stuck in a local minima and speed up convergeance. And finaly, the learning rate of 0.01 is pretty standard, but we don't want it to be too high or too low as it could cause losses to the convergeance and slow down the training.

Training history

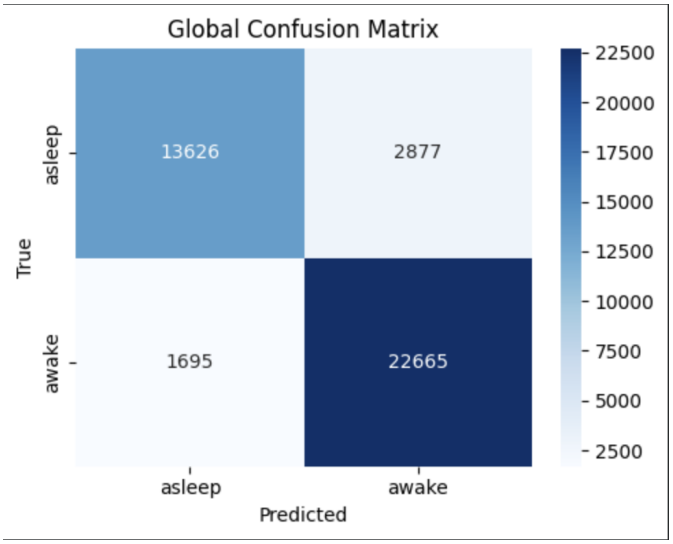


Both `train_loss` and `val_loss` drop pretty drastically ine the first two epochs and the decrease slowly and steadily at the same time. They converge around 0.084. There is no visible overfitting as the two curves remaing close during the training and `val_loss` never increases.

Performance

Global

Confusion matrix for each fold is available in the corresponding assetes folder.



The accuracy is as follows : (13626 + 22665) / 40863 = 88.8%. With this result, we can affirm that the model correctly classifies the large majority of samples in both classes. The error that is mostly present is when the model predicts `awake` when the mouse is actually `asleep` (2877 cases), it is not completly absurd as the light n-rem has EEG patterns that are similar to `awake` state. Here is the F1 score for each class and each fold:

Fold	F1 awake	F1 asleep	F1 micro
1	0.908	0.852	0.886
2	0.911	0.863	0.892
3	0.906	0.854	0.886

Second experiment

Pretreatment

Same as the first experiment, we chose the 25 lowest frequencies from 1Hz to 25Hz. The `StandardScaler` centers each feature to mean 0 and std 1. Since we now have 3 classes, the labels are encoded using `OneHotEncoder` which produces a one-hot vector per sample (e.g. `[1,0,0]` for rem, `[0,1,0]` for n-rem, `[0,0,1]` for awake).

Architecture

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 8)	208
dense_1 (Dense)	(None, 3)	27

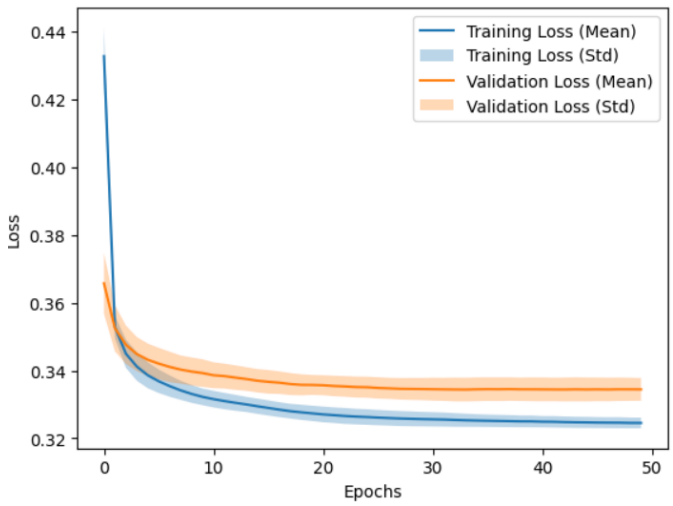
```
Total params: 235 (940.00 B)
Trainable params: 235 (940.00 B)
Non-trainable params: 0 (0.00 B)
```

Compared to the first experiment, the main changes are: - **3 output neurons** instead of 1, one per class (rem, n-rem, awake) - **Softmax** activation instead of sigmoid, each output represents a probability and the three sum to 1 - **Categorical crossentropy** loss instead of MSE, which is more appropriate for multi-class classification

The predicted class is determined by `argmax` of the output vector, which avoids the ambiguity of a threshold (with sigmoid or tanh, multiple classes or no class could be predicted simultaneously).

We kept the same single hidden layer with 8 neurons, the same learning rate of 0.001 and momentum of 0.99.

Training history

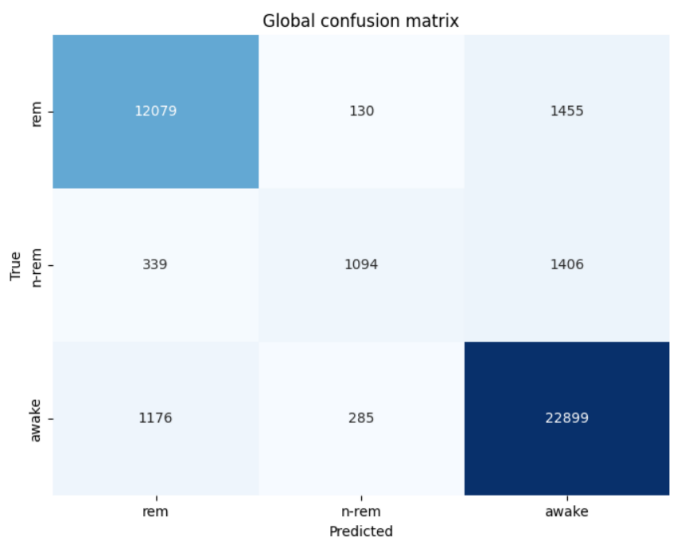


Both `train_loss` and `val_loss` drop sharply in the first few epochs then decrease slowly and steadily. They converge around 0.32-0.34. The validation loss is slightly higher than the training loss but remains close throughout training, indicating no significant overfitting. The std bands are narrow which shows consistent behaviour across the 3 folds.

Performance

Global

Confusion matrix for each fold is available in the corresponding assetes folder.



The model performs well on **rem** and **awake** which are the most represented classes. However, **n-rem** is the most confused class, out of 2839 total n-rem samples, only 1094 are correctly classified. It is frequently misclassified as awake (1406 cases) or rem (339 cases). This is expected as n-rem shares EEG characteristics with both other states, especially light n-rem which resembles the awake state.

Fold	F1 rem	F1 n-rem	F1 awake	F1 micro
1	0.887	0.513	0.914	0.883
2	0.884	0.509	0.912	0.881
3	0.888	0.487	0.915	0.884

The micro F1 score around **0.88** confirms that the model generalizes well on rem and awake, but struggles with n-rem which brings the overall score down.

Competition

Model Summary

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 8)	208
dense_1 (Dense)	(None, 3)	27

Total params: 235 (940.00 B)
Trainable params: 235 (940.00 B)
Non-trainable params: 0 (0.00 B)

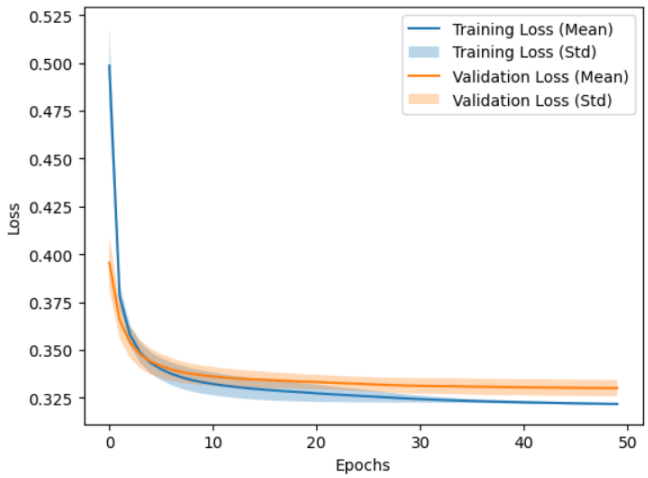
Layer	Details
Input	25 features (`amplitude_around_1_Hertz` ... `amplitude_around_25_Hertz`)
Dense (hidden)	8 neurons, ReLU activation : 208 parameters
Dense (output)	3 neurons, Softmax activation : 27 parameters
Total params	**235 (940 B)**

Chosen Hyper-parameters

Hyper-parameter	Value	Justification
Optimizer	**Adam**	Adaptive learning rate per parameter, faster convergence than SGD
Learning rate	0.001	Standard default for Adam
Hidden neurons	8	Same as Experiment 2 baseline
Activation (hidden)	ReLU	Avoids vanishing gradient
Activation (output)	Softmax	Outputs a probability distribution over 3 classes
Loss	Categorical Crossentropy	Standard loss for multi-class classification
Epochs	50	Sufficient for convergence with Adam
Folds	3	3-fold cross-validation as required

Implemented idea: replacing SGD (used in Experiment 2) with Adam. Adam maintains a running estimate of the first and second moments of the gradients, allowing it to adapt the learning rate individually for each parameter. This leads to faster and more stable convergence compared to SGD with a fixed step size.

Training History



Performance Results

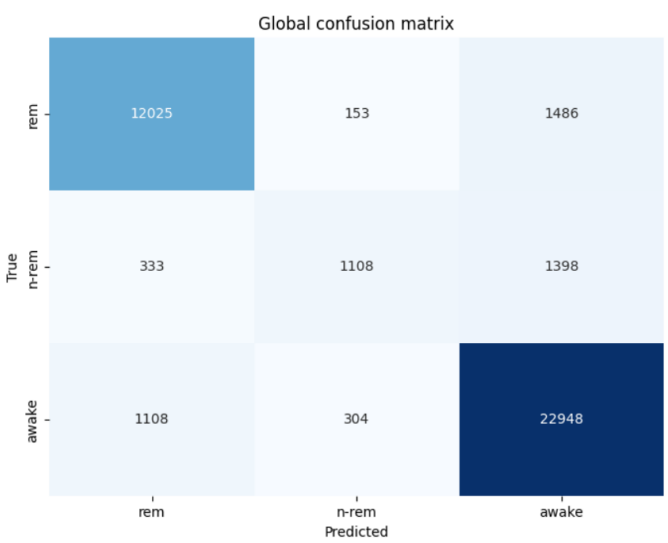
F1-score per class

Fold	REM	N-REM	Awake	Macro F1
Fold 1	0.8851	0.5024	0.9140	0.7672
Fold 2	0.8838	0.5236	0.9126	0.7733
Fold 3	0.8906	0.4824	0.9166	0.7632
Mean	**0.8865**	**0.5028**	**0.9144**	**0.7679**

Confusion matrices

Global

Confusion matrix for each fold is available in the corresponding assetes folder.



Analysis

The training and validation loss curves show fast and stable convergence — both losses drop sharply in the first 5 epochs and then plateau around 0.325 with very little gap between train and validation, indicating no significant overfitting.

The results across the three classes are however uneven:

- **Awake** is classified very well (F1 = 0.91), which is expected as it is the most represented class and has distinct EEG features.
- **REM** also performs well (F1 = 0.89), despite a non-negligible number of samples being confused with Awake (1486 globally).
- **N-REM** is the weakest class (F1 = 0.50). It is the least represented and its EEG signal overlaps with both REM and Awake, making it inherently harder to classify. The confusion matrix confirms this: N-REM samples are frequently predicted as Awake (1398 globally).

Compared to Experiment 2 which used SGD, Adam converges in fewer effective steps because it adapts the learning rate per parameter. This translates into a better-calibrated softmax output and a slightly higher macro F1-score.

Conclusion

By replacing SGD with Adam, a single, well-motivated change we obtained a mean macro F1-score of **0.7679** across 3 folds. The main limitation is the N-REM class, which remains difficult to classify due to class imbalance and spectral overlap with the other stages. The architecture and preprocessing pipeline are identical to Experiment 2, making the contribution of the optimizer change directly measurable.